



Haichao Zhang

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Education

University of Liverpool , Ph.D. in Artificial Intelligence and Large Language Models	2024.06 – Present
Research interests include large language models (LLMs), foundation models, retrieval-augmented generation (RAG), generative reasoning, and model optimization.	
Xi'an Jiaotong-Liverpool University , M.Sc. in Computer Science	2022.09 – 2024.03
Graduated with Distinction; ranked in the top 1% of the program.	
East China Jiaotong University , B.Eng. in Computer Science and Technology	2014.09 – 2018.06
Average score: 87/100; ranked in the top 1% of the program.	

Work Experience

Alibaba (Beijing) Software Technology Co., Ltd. , Development Engineer	2018.11 – 2022.08
- Developed data platforms for Alibaba's advertising delivery and recommendation systems, covering core components such as the audience engine, inspection and attribution, and tag computation. - Contributed to large-scale data platforms, complex business systems, and core engine platforms, with hands-on experience in high-concurrency system development and optimization for complex scenarios.	
Dingfu Data Technology (Beijing) Co., Ltd. , Development Engineer	2018.07 – 2018.11
Contributed to computer-vision algorithm development and image-processing projects, with a focus on image-to-table conversion and watermark removal.	

Academic Research

Customized RAG with Large Language Models and Retrieval-Augmented Generation	2025.01 – 2025.05
- First-author paper, Customized Retrieval-Augmented Generation with LLM for Debiasing Recommendation Unlearning, accepted by ICDM 2025 (CCF-B). - Proposed CRAGRU, which combines LLMs with retrieval-augmented generation (RAG) to support fine-grained knowledge retrieval, context construction, and generative reasoning; validated its effectiveness on multiple benchmark datasets. Project page: https://zhang-haichao.github.io/CRAGRU-Page - Technical focus: LLMs, RAG, retrieval augmentation, context modeling, and generative reasoning.	
Parameter-Efficient Model Adaptation with Prompt Tuning and Knowledge Distillation	2025.04 – 2025.08
- First-author paper, Teaching to Forget: Dual-Teacher Distilled Prompt-Tuning for Efficient Recommendation Unlearning, submitted to SIGIR 2026 (CCF-A). - Combined dual-teacher distillation with parameter-efficient prompt tuning to enable lightweight adjustment of targeted model behaviors without retraining the backbone model. - Technical focus: PEFT, prompt learning, knowledge distillation, and parameter-efficient adaptation.	
Controllable Generation with Inference-Time Control	2025.10 – 2026.01
- First-author paper, Controllable Generative Recommendation via Guided Token Refinement, submitted to IJCAI 2026. - Proposed ReGen, a controllable generation framework that reformulates generation as parallel iterative refinement over Semantic IDs, enabling real-time control during inference. - Technical focus: Transformers, generative modeling, inference-time control, parallel decoding, and constrained generation, balancing accuracy and diversity.	

Disentanglement

- First-author paper, Personalized Conformity Disentanglement for Debaised Recommendations.
- Proposed **PCDR**, which disentangles interest and conformity representations from dynamic user-behavior sequences to improve individual-level modeling.
- Technical focus: **Transformers, sequential modeling, attention, causal inference, and contrastive learning**; validated the method on multiple datasets.

Publications and Research Outputs

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- **Explain-then-Forget: Causal Explanation-based Unlearning for Efficient and Precise Recommendation**, ICDM 2026.
 - **Controllable Generative Recommendation via Guided Token Refinement**, ICDM 2026.
 - **Explaining Graph Recommendations via Counterfactual Support Sets**, ICONIP 2026.
 - **Topology-Aware Inverse Preference Learning for Robust Hybrid Voting Systems**, ICONIP 2026.
 - **Customized Retrieval-Augmented Generation with LLM for Debiasing Recommendation Unlearning**, ICDM 2025.
 - **Machine Vision-Enabled Octahedral Network Reconstruction and Structural Analysis of Perovskite Quantum Dots**, ACS Nano.
 - **Clustering-based Incremental Learning for Imbalanced Data Classification**, Knowledge-Based Systems.
 - **Uncertainty-Aware Semantic Decoding for LLM-Based Sequential Recommendation**, APWeb 2025.
 - **Counterfactual Contrastive Learning for Fine Grained Image Classification**, ICANN 2024.
 - **Two-branch Network with Feature Fusion for Time Since Deposition Estimation of Bloodstains**, CSCWD 2024.

Intellectual Property

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- **Invention Patent**: A Debiasing Method and Recommendation System Based on Personalized Causal Decomposition;
Publication No.: CN117312675A; **Publication Date:** December 29, 2023.
 - **Software Copyright**: Unbiased Recommendation Algorithm System Based on Personalized Conformity Disentanglement V1.0;
Registration No.: 2025SR0516629; **Registration Date:** March 25, 2025.
 - **Software Copyright**: Large-Scale Road-Network Taxi Dynamic Dispatch and Organization Platform for Mobile Internet Ridesharing; [YiBu V1.0]
Registration No.: 2018SR047738; **Registration Date:** January 22, 2018.
 - **Software Copyright**: Android-Based Railway Equipment Inspection System; [Railway Inspection Equipment APP V1.0]
Registration No.: 2018SR717297; **Registration Date:** September 6, 2018.
 - **Software Copyright**: Web-Based Railway Equipment Inspection System; [Railway Inspection Equipment System V1.0]
Registration No.: 2018SR643652; **Registration Date:** August 13, 2018.

Project Experience**Audience Engine Management Platform**

2020.04 – 2022.09

- Built unified management, workflow visualization, and anomaly-localization capabilities for the core audience engine of Alibaba's advertising data platform.
- Served as project lead, completing platform design, core feature development, and production delivery while coordinating interface design and platform iterations with engine owners.
- Established the system as the central management platform, significantly improving troubleshooting and operational efficiency.

Million-Scale Audience Inspection and Root-Cause Analysis Platform

2020.04 – 2022.09

- Built a closed-loop system for million-scale audience inspection, anomaly detection, root-cause analysis, and issue remediation across Alibaba's core advertising-delivery pipeline.
- Served as project lead, designing the system architecture and core framework while coordinating multiple module owners to deliver the end-to-end solution.
- Supported dozens of business lines after launch and reduced customer complaint rates by more than 90%.

- Designed the real-time tag-computation and audience-construction engine for advertising-delivery scenarios and developed its core modules.
- Optimized the tag-production pipeline using graph computation, covering the computation kernel, index design, DSL definitions, and other critical modules.
- Reduced tag iteration time from 7 days to 0.5 days and saved approximately 70% of computing resources.

Skills

- **Programming Languages:** Python (4 years), Java (6 years).
- **Deep Learning Frameworks:** Proficient in PyTorch; capable of independently completing model design, training, evaluation, and optimization.
- **Large Language Model Technologies:** Familiar with LLMs, RAG, Transformers, prompt tuning, knowledge distillation, model inference, and parameter editing.
- **Systems and Data Engineering:** Experience developing big-data and application systems with Flink, MapReduce, HBase, Spring Cloud, and MySQL.
- **Research:** End-to-end research experience spanning problem formulation, algorithm design, experimental validation, and academic writing.
- **Languages and Certifications:** LanguageCert IESOL C1 Pass; Intermediate Professional Title / Software Designer.

Honors and Awards

- First Prize, East China Jiaotong University ACM Programming Competition (2015.11).
- Provincial Second Prize, Jiangxi Computer Works Competition and “Internet+” Innovation and Entrepreneurship Competition (2016.06).
- First Prize, National Transportation Universities Outstanding Student Paper Competition (2016.09).
- National Third Prize, China University Transportation College Maker Competition and “Internet Transportation” Innovation and Entrepreneurship Competition (2016.11).

Professional Summary

- Combines industrial-scale system development with academic research, including experience building large-scale data platforms and high-concurrency systems at Alibaba as well as sustained research output in large language models.
- Research focuses on LLMs, RAG, controllable generation, knowledge editing, and parameter-efficient adaptation, with end-to-end capability from problem formulation and algorithm design to experimental validation.
- Strong programming foundation with the ability to independently drive complex projects, collaborate across teams, and deliver reliable results in demanding environments.